

Fritz – a supercomputer based on the same Fritz program that Kasparov played against. Kramnik managed to draw the tournament, billed as the “Brains in Bahrain”.

In 2006, Kramnik faced off against Fritz again and Fritz won. Many artificial intelligence experts have claimed this as the final victory of computers in the game of chess, with AI now being superior to the human brain in at least this game.

Robocup

While games like chess test the artificial intelligence of computers, a new competition among the world’s leading robotic developers is coming up with much more rapid strides in the field. The annual “RoboCup” competition puts a bunch of robots in a small football field to play the game.

“Football is a useful task for scientists developing robotic artificial intelligence because it requires the robot to perceive its environment, to use its sensors to build a model of that environment and then use that data to reason and take appropriate actions,” said researcher Claude Sammut who wrote a paper on the RoboCup in WIREs Cognitive Science in 2010. “On a football pitch that environment is rapidly changing and unpredictable requiring a robot to swiftly perceive, reason, act and interact accordingly.”

As with human players, football also demands communication and cooperation between robotic players and crucially requires the ability to learn, as teams adjust their tactics to better take on their opponents. And while the tiny humanoid NAO robots used for the RoboCup are impressive to look at, almost every developer agrees it’s the software – i.e. the artificial intelligence – that makes them so human-like.

Take, for example, the UT Austin Villa team that won the 2011 RoboCupSoccer championships in the 3-D simulation division. The key to victory, says team leader Patrick Stone, was he and his graduate and undergraduate students teaching their robots to teach themselves.

Most of the other teams used hand-coding, choosing where to put the joints, how to tweak the walk, etc. The UT Austin Villa team used a distributed computing cluster in order to do machine learning. Every night, the robots practiced. They would try walking one way and score themselves based on how quickly and how stably they could get to the ball and direct it toward the goal, and then the next night they’d do it again, trying different approaches and refining the ones that worked.